

Max Hamilton

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Education

University of Massachusetts, Amherst

Ph.D. in Computer Science

Advisor: Subhransu Maji

Sep 2022 - Dec 2026 (Expected)

University of Michigan, Ann Arbor

M.S.E. in Computer Science and Engineering

Cumulative GPA: 4.00/4.00

Sep 2020 - May 2022

University of Michigan, Ann Arbor

B.S.E. in Computer Science and Engineering

Minor in Mathematics

Cumulative GPA: 4.00/4.00

Sep 2016 - May 2020

Research Experience

Computer Vision Lab, Graduate Research Assistant, Amherst, MA

Advisor: Prof. Subhransu Maji

Sep 2022 - Present

- Developed *Active Measurement*, an adaptive sampling framework yielding statistically unbiased, high-precision aggregates from unlabeled data. Drastically reduced human labeling costs for quantifying malaria cells, satellite building damage, and bird populations.
- Extended unbiased estimators to compute Two-Point Correlation Functions (2PCF) for large-scale astronomy datasets.
- Created a zero-shot species range estimation model. Engineered a scalable multimodal pipeline processing 37k+ Wikipedia articles into embeddings to train a novel text-to-range-map system using observational and textual data.

Dolby Laboratories, PhD Research Intern, Sunnyvale, CA

Supervisor: Jaclyn Pytlarz

June 2024 – Sep 2024

- Researched and implemented efficient fine-tuning strategies for diffusion models on geometric vision tasks (depth and surface normal estimation)
- Matched state-of-the-art relative depth performance while reducing inference cost by $8\times$, enabling faster deployment in resource-constrained environments

Owens Research Group, Graduate Research Assistant, Ann Arbor, MI

Advisor: Prof. Andrew Owens

Apr 2021 - Sep 2022

- Developed correspondence-wise loss functions to achieve state-of-the-art results in video next frame prediction tasks
- Incorporated optical flow methods into loss function to compute correspondences between image predictions and ground truth

Fouhey AI Lab, Undergraduate Research Assistant, Ann Arbor, MI

Advisor: Prof. David Fouhey

May 2019 - Apr 2021

- Led project to perform weather pattern localization using machine learning in PyTorch.
- Built a large-scale dataset of 10M+ geo-tagged sky images to enable supervision at scale
- Explored self-supervised and implicit representations for image-based geo-localization

Vision and Learning Lab, Undergraduate Research Assistant, Ann Arbor, MI

Advisor: Prof. Jia Deng

May 2018 - Jan 2019

- Implemented a two-stage optimization pipeline in C++ to reconstruct 3D meshes from annotated 2D images
- Applied numerical optimization techniques to solve constrained curvature minimization for surface normal estimation
- Built a custom web crawler to collect 140,000+ real-world object images and extract camera pose from EXIF metadata

Publications

Scalable Model-Assisted Multi-Target Estimation in Large Image Collections

Max Hamilton, Subhransu Maji, Daniel Sheldon

In Submission, 2026.

Active Measurement of Two Point Correlations

Max Hamilton, Subhransu Maji, Daniel Sheldon

International Conference on Artificial Intelligence and Statistics (AISTats), 2026.

Active Measurement: Efficient Estimation at Scale

Max Hamilton*, Jinlin Lai*, Wenlong Zhao, Subhransu Maji, Daniel Sheldon

Advances in Neural Information Processing Systems (NeurIPS), 2025.

Feedforward Few-shot Species Range Estimation

Christian Lange, **Max Hamilton**, Elijah Cole, Alexander Shepard, Samuel Heinrich, Angela Zhu, Subhransu Maji, Grant Van Horn, and Oisin Mac Aodha

Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.

Combining Observational Data and Language for Species Range Estimation

Max Hamilton, Christian Lange, Elijah Cole, Alexander Shepard, Samuel Heinrich, Oisin Mac Aodha, Grant Van Horn, and Subhransu Maji

Advances in Neural Information Processing Systems (NeurIPS), 2024.

Comparing Correspondences: Video Prediction with Correspondence-wise Losses

Daniel Geng, **Max Hamilton**, and Andrew Owens

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022.

OASIS: A Large-Scale Dataset for Single Image 3D in the Wild

Weifeng Chen, Shengyi Qian, David Fan, Noriyuki Kojima, **Max Hamilton**, Jia Deng

IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2020.

Teaching Experience

University of Massachusetts Amherst, Teaching Assistant

CompSci 670: Computer Vision

Spring 2024

CompSci 682: Neural Networks

Fall 2023, Spring 2026

University of Michigan, Graduate Student Instructor

EECS 542: Advanced Topics in Computer Vision

Winter 2022

Honors & Awards

Edward Riseman and Allen Hanson Scholarship, University of Massachusetts, Amherst

Apr 2023

Christopher P. Moeller Scholarship Award, University of Michigan

Sep 2021

Awarded to outstanding graduate students studying computer science or mathematics.

EECS Scholar, University of Michigan

2019 - 2020

Tau Beta Pi Honor Society, University of Michigan

Apr 2018

Dean's List, University of Michigan

2016 - 2020

Professional Activities

Mentor, Early Research Scholars Program, University of Massachusetts, Amherst 2024 – Present
Program committee, New England Computer Vision (NECV) Workshop 2025
Reviewer: ICCV 2023, WACV 2024, ECCV 2024, WACV 2025, NeurIPS 2025, CVPR 2026

Skills

Programming Languages: Python, SQL, C++, MATLAB, Java
Machine Learning: PyTorch, PyTorch Lightning, NumPy, Hugging Face, TensorBoard, Optuna
Distributed Computing: Amazon Web Services (AWS), Google Cloud Platform (GCP), Slurm